

YASHASWI A R

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github.com/yashaswiar-svg

SUMMARY

Motivated MSc Data Science graduate with a strong foundation in machine learning, statistical modeling, and data-driven problem solving. Proficient in Python, SQL, and key ML frameworks, with hands-on experience through academic projects and generative AI development. Passionate about transforming complex datasets into meaningful insights that drive business decisions. Seeking an entry-level Data Scientist role to apply analytical skills and contribute to impactful, real-world solutions.

TECHNICAL SKILLS

Programming: Python, R, HTML/CSS (basic)

Databases: MySQL, PostgreSQL, MongoDB (basic)

Data Tools: Pandas, NumPy, Excel

Data Visualization: Tableau, Power BI

Machine Learning: Supervised (Regression, Decision Trees, Random Forest, SVM) · Unsupervised (K-Means, DBSCAN, PCA, t-SNE) · Ensemble (XGBoost, LightGBM, Gradient Boosting) · Neural Networks (ANN, CNN, RNN, LSTM)

Data Mining: Association Rule Mining, Clustering, Anomaly Detection, Dimensionality Reduction (PCA, t-SNE), Pattern Recognition

Statistics & Mathematics: Inferential & Descriptive Statistics, Probability Theory, Regression Analysis, Hypothesis Testing, A/B Testing, Bayesian Statistics

Concepts: EDA, Feature Engineering, NLP, Statistical Modelling, A/B Testing

EDUCATION

M.Sc. – Data Science

Chanakya University — Bangalore, India 2025 – Present

B.Sc. – Computer Science & Statistics

St. Agnes College, Mangalore University — Mangalore, Karnataka Aug 2022 – Jun 2025

CGPA: 8.45

RESEARCH PUBLICATION

Student Performance Prediction Using K-Nearest Neighbor and Comparative Machine Learning Algorithms: An Extended Empirical Study

Extended Research Article | Tools: Python, scikit-learn, pandas, NumPy, GridSearchCV, Kaggle Dataset 2025

- Extended the foundational KNN study by Suyal & Goyal (2022) with a full empirical pipeline on 1,000 real student records (Kaggle), addressing limitations of small dataset size and absence of comparative analysis.
- Engineered a Pass/Fail binary label from average subject scores; applied StandardScaler, 80/20 train-test split, and Stratified K-Fold cross-validation to handle class imbalance (~940 Pass vs ~60 Fail).
- Conducted systematic hyperparameter tuning via 5-fold GridSearchCV across $k \in \{1-20\}$ and three distance metrics; identified optimal KNN configuration ($k=13$, Euclidean) achieving 100% test accuracy and 99.25% CV accuracy.
- Benchmarked KNN against Logistic Regression, Decision Tree, Random Forest (99.5%), and SVM — all achieving near-perfect accuracy, confirming linear separability of the feature space.
- Authored a full research paper covering introduction, related work, methodology, results, discussion, and future scope including SMOTE, SHAP explainability, and multi-class extensions.

PROJECTS

AI Dashboard — Conversational BI for E-Commerce

Next.js, FastAPI, Python, Google Gemini, Firebase, Docker 2026

- Built an AI-powered BI platform using a 3-step agentic pipeline (Plan → Generate SQL → Validate & Execute) with Google Gemini LLM, converting natural-language questions into 9 interactive Plotly dashboards.
- Integrated Firebase Authentication, multi-format export (PDF, JSON, CSV), and SQLite-backed data ingestion (CSV, XLSX, Amazon live data) for hallucination-free query results.

Fact & Claim Verification System

Python, FastAPI, Google Gemini, Web Search API 2026

- Built an AI-powered fact-checking engine that extracts claims from news articles (URL or raw text) and classifies each as True / False / Partially True / Unverifiable with citations and confidence scores.
- Engineered a claim extraction pipeline (LLM structured JSON output) with an evidence retrieval module that autonomously cross-references real-world sources, plus AI-content detection and mock mode for offline testing.

Mitra-NammaEco — AI Financial Mitra

Node.js, Firebase, Google Gemini Pro, WhatsApp API 2026

- Built a voice-first AI financial assistant on WhatsApp for informal workers in Bengaluru, supporting 7 languages with Gemini AI extracting income/expense data from natural speech into Firebase Firestore.
- Implemented government scheme matching (Ayushman Bharat, PM-SVANidhi), crisis-aware empathy mode, and break-even market math to help street vendors and auto drivers make smarter financial decisions.

CERTIFICATIONS

- Machine Learning Certificate — IntrnForte (2025)
- Python Programming — IntrnForte (2025)